

ACADOS-MATLAB WORKFLOW

“complete” model

“dual_arm_reduced_xy_dynamics.m”:

This script is used to symbolically derive the B matrix and n vector of the complete dynamics. The algorithm is based on Screw-Theory formulation, and it is described in [1].

Running this script produces B and n as symbolic variables (using MATLAB symbolic toolbox).

Note: there is no direct conversion between matlab symbolic variables and casadi SX variables. The components of B and n can be read as a strings and copy-pasted into the following model.m script

“dual_arm_reduced_xy_model.m”:

This script is actually a matlab function that outputs a “model” data structure used to define the optimal control problem (OCP).

The function defines the numerical values of the derived model, then declares the model used for the OCP. The model is written in joint acceleration-input form [2]. The constraint on the arm distance is also included into the output data structure.

“dual_arm_reduced_xy_ocp.m”:

This script effect can be derived in 3 parts:

- **NMPC problem definition**

- Set the NMPC parameters (sample time, shooting nodes, prediction horizon etc.)
- Declare the models for OCP and the plant
- Set the weight in the cost function
- Set the NMPC constraints
- Set the numerical optimization parameters (solver type, tolerance, etc.)

- **Code generation**

- Generate the C code, mex files and matlab scripts. The majority of the files are mex files and matlab scripts and are needed in order to run the simulation of the controller and the plant on MATLAB of the next section.
- Also the object files (ending in .s) and shared object files (ending in .so) can be ignored
- The files needed to be included into the CRF are the function definition files (ending in .c) and the header files (ending in .h) **for the solver only**. Files containing “sim” and/or “plant” in the title refer to the model of the plant and are not needed into the CRF.

- **Simulation**

- Iteratively solves the OCP defined above
- Takes the first input of the optimal input sequence and feeds it to the plant model
- Takes the output of the plant model and uses it to initialize the next OCP

***additional note:** in the file titles and in the code (both matlab and generated c code) the word “reduced” is used referencing the complete model. This is because with the old nomenclature, the word “complete” used to reference the general cable-suspended-robot model stated in [3], while the “reduced” version is the one taking less degrees of freedom after the simplifying hypothesis [2]. In order to improve clarity, the word should be renamed in the future.

“mass-spring-damper” model

“mass_spring_damper_model.m”:

This script is actually a matlab function that outputs a “model” data structure used to define the optimal control problem (OCP).

The function defines the numerical values of the derived model, then declares the model used for the OCP. The model is in section 4.2.2 of [2].

“mass_spring_damper_ocp.m”:

This script is completely analogous to

“dual_arm_reduced_xy_ocp.m” but uses the mass-spring-damper model instead.

“dual_arm_reduced_xy_model.m”:

This is the same file of the complete model as seen in the previous section. It is only used to simulate the plant within the MATLAB script, any code generated by this should be ignored.

References [todo: add links]:

- [1] Giancarlo's Thesis
- [2] Michele's Thesis
- [3] Giancarlo's Paper
- [4] Acados documentation